Dirac-before-Routh Constrained Spectra in Bianchi Class-A Einstein–Cartan–Nieh–Yan Cosmologies: Rank-Defect Kernels, Spectral Robustness, and Torsion-Sector Closure

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Abstract

We analyse the Lorentzian first-order Einstein–Cartan plus Nieh–Yan (EC+NY) system on Bianchi class-A backgrounds using a Dirac-before-Routh (DBR) constraint-reduction protocol. The rank defect of the Bianchi class-A structure matrix is a mathematically known invariant; our result is that this defect is realised, in the DBR-standard quadratic constrained perturbation spectrum, as a residual non-propagating scalar kernel. Across the four backgrounds T^3, Nil^3, Sol^3, and S^3, the propagating tensor sector contains two positive-kinetic transverse-traceless (TT) graviton degrees of freedom, and the dimension of the scalar kernel is

$$d_{\mathrm{zero}} = \dim\ker n = 3 - \mathrm{rank}(n).$$

This classification is established through the DBR constraint algebra, a frame-invariance check, scalar/tensor spectral analysis, and a scoped $k > 0$ spectral completion over the analysed harmonic sectors. Reversing the order of operations – eliminating the auxiliary variables before Dirac closure – produces a Dirac-inequivalent extra scalar mode in the verified S^3 scalar sector; in light of this control result, all subsequent physical claims are made within the DBR protocol. We further classify the finite deformations of the zero kernel into exact moduli and curvature walls within a finite Palatini branch, and we show that the torsion ansatz scalars remain an exact auxiliary under the NY-MAIN route, acquiring no kinetic or gradient terms even under field promotion, so that the torsion sector does not affect the above hierarchy of results. All results are established within the explicitly stated quadratic, harmonic-sector, and homogeneous scopes.

1 Introduction

1.1 Background and motivation

The DPPU series has progressively organised the Einstein–Cartan plus Nieh–Yan setting [8, 9] in terms of homogeneous geometry, torsion modes, and topology response [1–5]. Paper V [6] constructed the Lorentzian $\alpha = 0$ EC+NY baseline on the S^3 closed-FLRW branch, and paper VI [7] extended it to the isotropic reduced sector of the four topologies $S^3/T^3/\mathrm{Nil}^3/\mathrm{Sol}^3$, establishing χ-controlled topology universality.

Those works analysed the isotropic reduced sector. The present paper takes the next step: it analyses the *homogeneous anisotropic perturbation spectrum of the same Lorentzian EC+NY system*, on the four Bianchi class-A backgrounds, keeping *first-order (Palatini-type) variables*. What becomes essential here is the question of the reduction route. In first-order systems it is known in general that the *order of operations* – whether one eliminates the auxiliary connection variables by their equations of motion (Routh-type elimination) before performing the constraint analysis, or completes the Dirac–Bergmann closure first and eliminates afterwards – can affect the result [13, 21–23]. We adopt the **Dirac-before-Routh (DBR)** protocol, which performs the Dirac closure first (Section 3), and classify the constrained perturbation spectra of the four topologies under this single common protocol.

1.2 Known results and our contribution

To make the scope of the novelty precise, we distinguish the known results from the contribution of this paper.

Known results.

- Dirac's constrained Hamiltonian analysis is a standard framework [10–12].
- Solving and substituting variables before constraint closure can change the constrained system unless equivalence is proved [13, 21, 22].
- Bianchi class-A Lie algebras are classified by the rank/signature data of their structure matrix [14–16].
- For the four topologies treated here, the rank-defect / abelianisation / b_1 patterns are geometrically known [16, 18–20].
- The coincidence $d_{\mathrm{zero}} = b_1$ ($\dim\ker n$ equals the first Betti number) is a known algebraic fact [16, 18].

Our contribution.

- In the Lorentzian EC+NY first-order Palatini setting of the DPPU framework, the DBR constrained perturbation spectrum realises the known Bianchi rank-defect as a residual non-propagating scalar kernel.
- The four-topology comparison is carried out under a single common DBR protocol, with universal propagating content (2 TT gravitons, positive kinetic) and only the non-propagating scalar kernel being topology-dependent.
- The reversed-order route (Routh-before-Dirac) is retained as a control: it produces a route-dependent Dirac-inequivalent extra mode in the verified S^3 scalar sector, and is therefore not used for physical claims.
- A scoped $k > 0$ spectral completion over the analysed harmonic sectors, a classification of the finite deformations of the zero kernel, and the torsion/NY verification together show that the DBR rank-defect kernel is not (within the stated scope) an artefact of the homogeneous quadratic truncation.

1.3 Main results

The main results of this paper are stated below in hierarchical order.

1. **Realisation of the rank defect.** In the DBR-standard quadratic constrained spectrum, the residual non-propagating scalar kernel realises the known Bianchi class-A rank defect on the physical-degrees-of-freedom side. The hierarchy is S^3:0 / Sol^3:1 / Nil^3:2 / T^3:3 (Bianchi IX/VI_0/II/I, $\mathrm{rank}(n) = 3/2/1/0$). The dimension is SO(3) frame-invariant, and the mechanism is derived from the activation rank of the DBR constraint algebra (Section 4):

$$d_{\mathrm{zero}} = \dim\ker n = 3 - \mathrm{rank}(n), \qquad n^{ij} = \tfrac{1}{2}\,\epsilon^{ikl}C^{j}{}_{kl}. \tag{1}$$

2. **Universality of the propagating content.** In all four topologies the propagating degrees of freedom are two TT gravitons, and the kinetic block is positive definite.
3. **Robustness from spectral completion.** The classification persists under a scoped $k > 0$ spectral completion over the analysed harmonic sectors (T^3/S^3 endpoint controls, Nil^3 closed-form completion, Sol^3 bulk gap plus positive compact edge; Section 5).
4. **Realisation of the zero kernel.** The zero kernel is realised as left-invariant harmonic one-forms ($\ker n$; Section 6).

5. **Finite fate of the zero kernel (supporting result).** The finite deformations of the zero kernel are classified, within the finite Palatini branch / homogeneous minisuperspace, into exact moduli and curvature walls: modulus/wall = T^3: 3/0, Nil^3: 1/1, Sol^3: 0/1, S^3: 0/0 (Section 7).

6. **Closure of the torsion sector (ansatz-restricted).** Within the NY-MAIN route and the native torsion ansatz scalars (η, V), the reduced torsion sector remains an exact auxiliary, and its promotion to fields $\eta(x), V(x)$ generates no quadratic kinetic/gradient term under the inspected native-action criterion. Hence torsion/NY does not affect the main results of Sections 4–7 (Section 8).

7. **Route comparison (control).** The route that performs the Routh elimination before Dirac closure produces a Dirac-inequivalent extra scalar mode in the verified S^3 scalar sector. It is recorded as a control route (Section 3, Appendix C) and is not used for physical claims.

Throughout, d_{zero} carries the following qualifications: (i) it is not claimed as a new topological invariant; (ii) it is not identified with a propagating physical zero mode; (iii) it is a residual non-propagating scalar kernel in the DBR quadratic constrained spectrum.

1.4 Claim hierarchy and scope

The claims of this paper have the following layered structure, with each layer carrying a different scope (domain of validity).

Layer	Content	Scope (domain of validity)
L0	Quadratic core theorem (results 1, 2, 4)	Within the DBR quadratic constrained spectrum (theorem)
L1	$k > 0$ spectral completion (result 3)	Analysed harmonic sectors (Sol^3 compact edge is representative numerics)
L2	Finite deformation of the zero kernel (result 5)	Finite Palatini branch / homogeneous minisuperspace (supporting result)
L3	Closure of the torsion sector (result 6)	NY-MAIN route + ansatz scalars + inspected criterion
R	Route comparison (result 7)	Verified S^3 scalar sector (control)

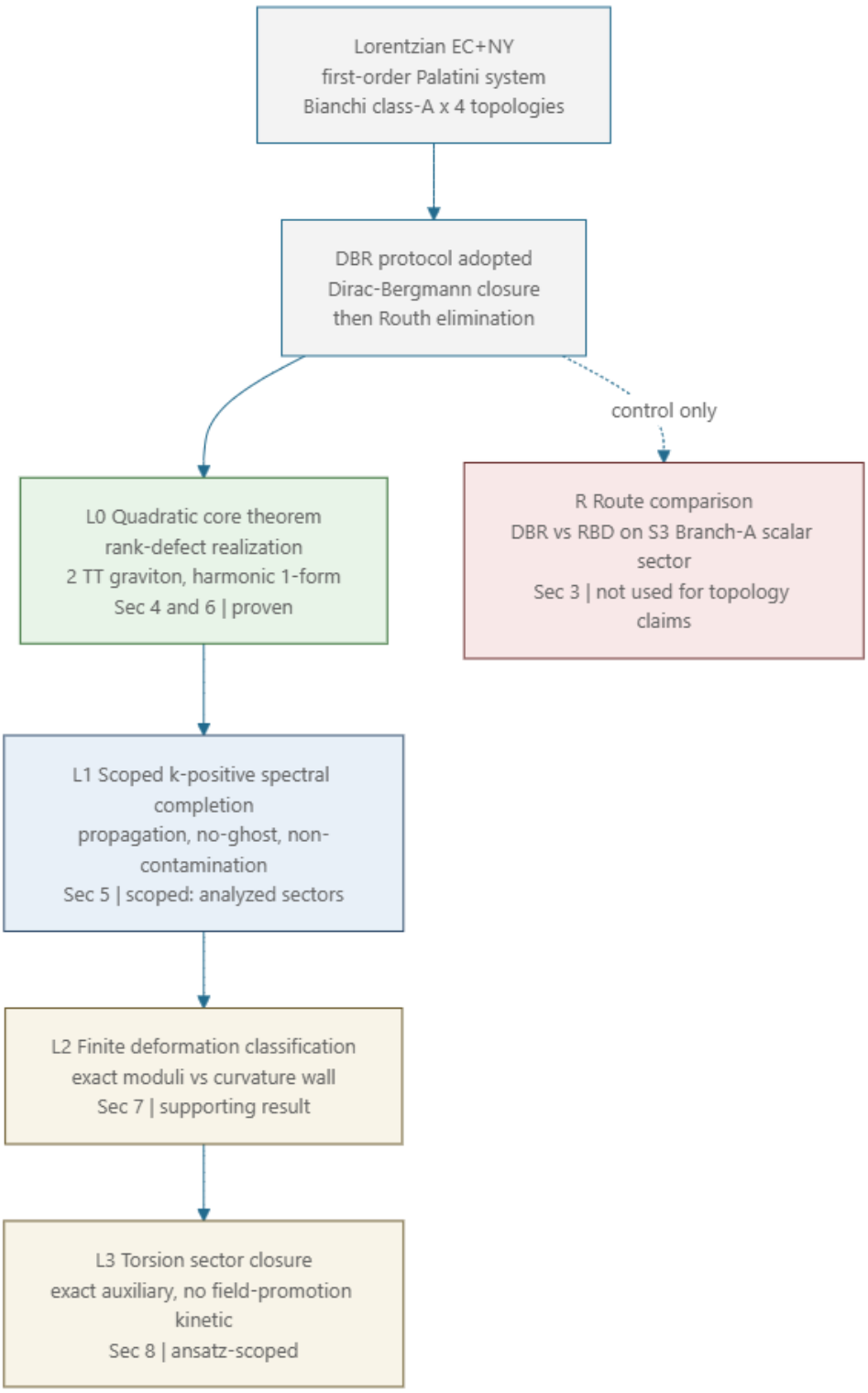


Figure 1: Claim hierarchy and DBR workflow. The DBR protocol is applied to the Lorentzian EC+NY first-order Palatini system, and the main results are built up in the layers L0→L1→L2→L3. All physical claims of L0–L3 are established under DBR. The route comparison R is the control basis for adopting DBR and is not used for the subsequent topology claims. *Scope*: the section numbers in the figure are reader-order pointers to the text; L2, L3, and R are scope-restricted as in the table above.

This paper does not claim a full nonlinear field-theory theorem, a complete PDE closure for arbitrary inhomogeneous perturbations, or a theorem on torsional charge/localisation. These boundaries are made explicit in the scope-and-limitations statements of each section and of Appendices C–G.

1.5 Relation to earlier DPPU papers and adjacent literature

In the isotropic reduced sector of paper VI [7], the four topologies did not change the admissibility pattern, and the topology dependence localised in the scalar χ. At the anisotropic perturbation level of the present paper, the topology dependence localises in the dimension d_{zero} of the non-propagating scalar kernel, whose value is the rank defect of the structure matrix. The two realise the same structure – "topology dependence localises in algebraic data" – in different sectors.

Relation to adjacent literature: the count of global degrees of freedom (Teichmüller parameters) of compact Bianchi models was established by Koike–Tanimoto–Hosoya [19, 24–26]. Our d_{zero} is *not the global moduli of metric GR*; it is a perturbative kernel in the DBR constrained spectrum of the first-order EC+NY system, a distinct object (Section 7 shows the finite moduli count $\neq d_{\text{zero}}$). The relation between the two is discussed in Section 9. Among existing studies of Bianchi perturbation theory, the gauge-invariant perturbation theory of Bianchi I is available [27]. The relation between the "constraint bifurcation / field activation" phenomena in the constraint analysis of Poincaré gauge theory [30, 31] and our activation rank is distinguished in Section 4.3. That the NY term is topological in the canonical analysis is known [33–35]; our torsion section establishes this exactly in the reduced system.

1.6 Organisation of the paper

Section 2 fixes the Lorentzian EC+NY setup and the Bianchi class-A backgrounds, the structure matrix, and the torsion ansatz. Section 3 defines the DBR reduction protocol and the route justification (summary of the control experiment). Section 4 states the central theorem: the rank-defect realisation in the DBR constrained spectrum. Section 5 gives Robustness I, the scoped $k > 0$ spectral completion (propagation, no-ghost, non-contamination). Section 6 gives the harmonic one-form realisation of the zero kernel. Section 7 classifies the finite fate of the zero kernel (exact moduli versus curvature walls). Section 8 establishes the torsion/Nieh–Yan sector robustness. Section 9 discusses the results and the outlook. Appendix A fixes conventions; Appendix B collects the native local action and DBR computational checks; Appendix C details the route comparison; Appendix D collects the endpoint and spectral evidence; Appendix E the harmonic-form and Hessian-kernel realisation; Appendix F the finite-fate branch data; and Appendix G the torsion auxiliary exactness and field-promotion verification.

2 Lorentzian EC+NY setup and Bianchi class-A backgrounds

2.1 First-order action ingredients

The basic variables are the tetrad one-form e^I and the Lorentz connection one-form ω^{IJ}, both treated as independent (first-order / Palatini type). The action consists of the Einstein–Cartan term [8] and the Nieh–Yan term [9]:

$$S[e,\omega] = \frac{1}{2\kappa^2}\int \epsilon_{IJKL}\, e^I \wedge e^J \wedge R^{KL}[\omega] + \frac{\theta_{\mathrm{NY}}}{\kappa^2}\int \left(T^I \wedge T_I - e_I \wedge e_J \wedge R^{IJ}\right), \qquad (2)$$

where $R^{IJ}[\omega] = \mathrm{d}\omega^{IJ} + \omega^I{}_K \wedge \omega^{KJ}$ and $T^I = \mathrm{d}e^I + \omega^I{}_J \wedge e^J$. The signature and the Hodge/ϵ conventions are fixed in Appendix A.

Treatment of the NY term. We adopt the route that treats the NY term as an exact four-form (the NY-MAIN route). Within our NY-MAIN route and the inspected reduced/native-action gate of the LZ-native AX/VT/MX torsion ansatz scalars, the NY term contributes on the background only as an exact boundary term. In the reduced system it forms an exact auxiliary shell, and its promotion to fields $\eta(x), V(x)$ generates no quadratic kinetic/gradient term under the inspected native-action criterion (Section 8). The topological nature of the NY term in the canonical analysis is known [33–35]. This statement of bulk non-involvement is restricted to the above scope and does not extend automatically to a full torsion field theory or to a general NY action.

2.2 Bianchi class-A backgrounds and the structure matrix

The spatial sections carry the left-invariant structure of a three-dimensional unimodular Lie group G. The left-invariant one-forms σ^i $(i = 1, 2, 3)$ satisfy

$$\mathrm{d}\sigma^i = -\tfrac{1}{2}\, C^i{}_{jk}\, \sigma^j \wedge \sigma^k. \qquad (3)$$

The class-A (unimodular) condition is $C^j{}_{ij} = 0$ [15], and then the structure constants are encoded in a symmetric *structure matrix* n^{ij}:

$$C^i{}_{jk} = \epsilon_{jkl}\, n^{li}, \qquad n^{ij} = \tfrac{1}{2}\,\epsilon^{ikl}\, C^j{}_{kl}, \qquad n^{ij} = n^{ji}. \qquad (4)$$

By the Milnor frame [16], $n^{ij} = \mathrm{diag}(n_1, n_2, n_3)$, and its rank/signature classifies the class-A type. The four topologies treated here are collected in Table 1. Bianchi models with compact spatial sections are restricted to class A [19, 20].

Topology	Bianchi	n (Milnor frame)	rank(n)	$d_{\text{zero}} = 3 - \text{rank}(n)$	b_1	$R\,q^2$
T^3	I	$\text{diag}(0,0,0)$	0	3	3	0
Nil^3	II	$\text{diag}(1,0,0)$	1	2	2	$-1/2$
Sol^3	VI_0	$\text{diag}(1,-1,0)$	2	1	1	-2
S^3	IX	$\text{diag}(1,1,1)$	3	0	0	6

Table 1: Four Bianchi class-A backgrounds. Here b_1 is the first Betti number of the corresponding compact quotient (standard values [18]), and $R\,q^2$ is the dimensionless spatial scalar curvature at the isotropic point.

The S^3 structure-constant normalisation is fixed to $C = \epsilon$ ($|C| = 1$) throughout (Appendix A, where the explicit conversion to the normalisation of paper VI [7] is also recorded). The quantities rank(n) and d_{zero} are invariant under an overall structure-constant scaling $C \to sC$, whereas the *numerical* values such as $R\,q^2$ and the TT eigenvalues depend on s (Appendix A.3).

The d_{zero} in Table 1 is the algebraic prediction from the rank defect of the structure matrix. That this is realised as the kernel of the constrained spectrum is the main result of this paper (Section 4). The relation to abelianisation, $\dim(\mathfrak{g}/[\mathfrak{g},\mathfrak{g}]) = \dim\ker n$, is also a known algebraic fact and is used only as a geometric cross-check.

2.3 Background ansatz and perturbation sectors

The background is taken as the homogeneous diagonal ansatz

$$e^0 = N(t)\,\mathrm{d}t, \qquad e^a = q_a(t)\,\sigma^a \quad (a = 1,2,3;\ \text{no sum}), \tag{5}$$

and the connection ω^{IJ} has independent homogeneous components (the explicit component list and the reduced action are in Appendix B). Relative to the background frame the perturbations decompose into

- a **scalar (diagonal/axis) sector**: diagonal metric perturbations β_a and the corresponding connection perturbations,
- a **spin-1 (transverse vector) sector**,
- a **spin-2 (TT) sector**: transverse-traceless perturbations h^{TT}_{ab}.

The analysis of the homogeneous modes ($k = 0$) is central to Section 4, while the inhomogeneous modes are treated by a topology-dependent harmonic expansion (T^3: Fourier; S^3: hyperspherical harmonics [28, 29]; Nil^3: toroidal $\cup$ Heisenberg sector; Sol^3: boost sector) in Section 5 and Appendix D.

2.4 Torsion ansatz scalars

On the homogeneous background, the torsion within the ansatz used here organises into the following scalar modes:

- **AX (axial)**: an axial scalar $\eta(t)$ giving $T^a{}_{bc} \supset \eta\,\epsilon_{abc}$;
- **VT (vector/trace)**: a scalar $V(t)$ giving the trace part;
- **MX (mixed)**: the mixed branch of the two.

These are the same family as the branch classification (EH/AX/VT/MX) of papers V/VI [6, 7]. Section 8 classifies the derivative couplings obtained when these are promoted to fields $\eta(x), V(x)$. All torsion claims of this paper lie *within the* AX/VT/MX *ansatz scalars*; no generalisation to a full 24-component torsion field theory is claimed.

2.5 Notation summary

In what follows, q^2 denotes the square of the isotropic scale factor (the isotropic point $q_1 = q_2 = q_3 = q$), $\kappa^2 = 8\pi G$, and vol denotes the comoving volume of the spatial cell. The notation of the constraint analysis (primary/secondary constraints, first/second class, Dirac bracket) follows the standard definitions [10–12].

3 DBR reduction protocol and route justification

3.1 The Dirac-before-Routh protocol

The phase space of the first-order EC+NY system consists of the tetrad, the connection, and their conjugate momenta, and it carries many primary constraints. Our reduction procedure adopts the following order *as a protocol*.

> **DBR (Dirac-before-Routh) protocol.** (i) Complete the Dirac–Bergmann algorithm [10–12]: generate the secondary constraints from the consistency of the primary constraints, and after closure classify first/second class and construct the Dirac bracket for the second-class pairs. (ii) Only then perform the Routh-type elimination / reduction of the auxiliary variables (such as the non-dynamical connection components) and present the reduced spectrum.

DBR is not a new principle; it is the standard procedure of constrained dynamics applied faithfully to this system. The reverse order – solving the auxiliary connection variables by their field equations and substituting into the action before the constraint analysis – is called the **RBD (Routh-before-Dirac)** route here.

3.2 Why the order matters: a recognised issue

That substitution at the level of the action is not, in general, interchangeable with substitution at the level of the equations of motion is an established generality [13]. Substitution is safe only for auxiliary variables (whose own equations are algebraically solvable and free of derivatives); otherwise a proof of equivalence is required. In the Bianchi context it is also classically known that the validity of imposing the symmetry at the level of the action is restricted to class A (in class B it gives the wrong field equations) [21, 22], and the equivalence of the Palatini and metric formulations itself is a nontrivial statement to be proved by constraint analysis [23]. In adopting DBR as the standard, we select the order of operations for which constraint equivalence is guaranteed against this known "Palatini reduction non-commutativity" (route dependence).

3.3 Control experiment

To support the adoption of DBR empirically, we ran the RBD route as a **control experiment**. The verification was carried out in the Branch-A scalar sector of the S^3 background. The result is the following.

> In the verified S^3 Branch-A scalar sector, eliminating the auxiliary connection variables before the Dirac closure produces a Dirac-inequivalent extra scalar

mode. The recomputation gives $N_{\text{phys}} = 0$ for DBR_KEEP_ALL at all representative harmonic levels, whereas the spatial-ω and full-K Routh variants give $N_{\text{phys}} = 1/1/2$ for $l = 0, 1, \geq 2$. At the lowest harmonic level the comparison starts from the same Branch-A local first-order Lagrangian, and reproduces the discrepancy DBR:0 / RBD:1 together with $H_{PP} = -3/8 < 0$. For $l \geq 1$ the RBD extra count is treated as a route-level gauge-breaking artefact; the negative-sign diagnostic is claimed only at $l = 0$ (scope details in Appendix C).

This discrepancy is *route-level*, not a physical instability. We do not draw from it any theorem that "RBD is generally wrong." The claim is confined to a single point: *since route equivalence demonstrably fails in this system, all subsequent physical topology claims are made within the DBR protocol.*

Route	Order of closure and elimination	Outcome in S^3 Branch-A scalar sector	Status
DBR	Dirac closure → Routh elimination	0 physical scalars (kernel non-propagating; Section 4)	**adopted** (standard)
RBD	Routh elimination → Dirac closure	Dirac-inequivalent extra scalar mode ($H_{PP} = -3/8 < 0$ at $l = 0$; route-level)	control route (recorded)

Table 2: Route taxonomy (compact version; details in Appendix C).

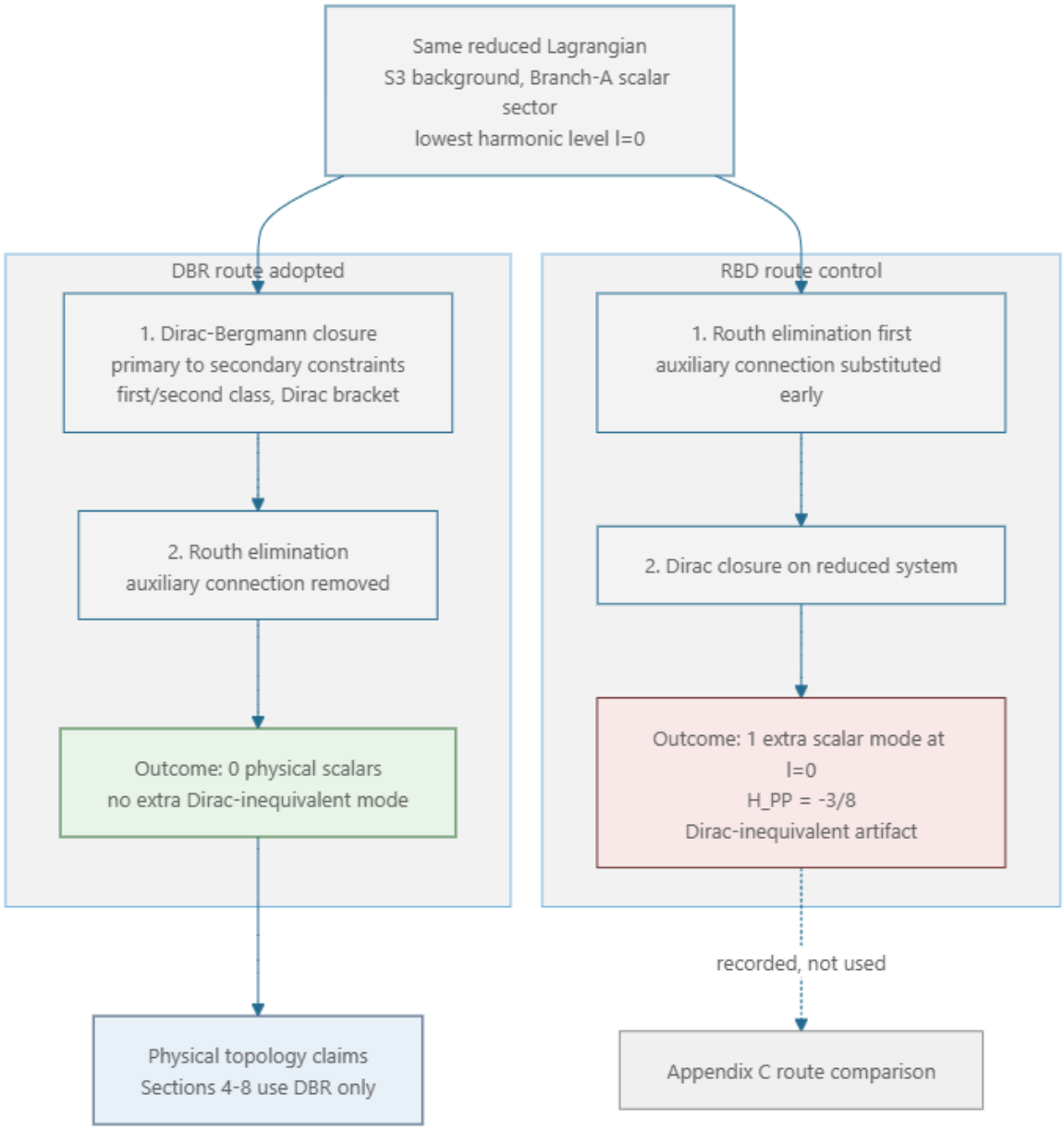


Figure 2: DBR versus RBD route comparison. From the same Branch-A local first-order Lagrangian, the two routes – differing only in the order of operations – are shown side by side. DBR yields 0 physical scalar degrees of freedom, while RBD gives a Dirac-inequivalent extra mode; at $l = 0$ the diagnostic sign is $H_{PP} = -3/8 < 0$. *Scope*: the negative sign is claimed only for the lowest level; for $l \geq 1$ the RBD extra count is recorded as a route-level gauge-breaking artefact. The RBD outcome is not a physical instability.

3.4 Scope declaration

The claims of all subsequent sections (Sections 4–8) are results under the DBR protocol. The RBD route is referred to only in the route-comparison details of Appendix C.

4 Rank-defect realisation in the DBR constrained spectrum

4.1 Theorem statement

The rank defect $\dim\ker n$ is itself a known invariant of the Bianchi class-A algebra [15, 16]. The following theorem concerns its *realisation*: it states that the scalar kernel surviving in the DBR-standard EC+NY constrained perturbation spectrum is given precisely by this known invariant.

Theorem 1 (Rank-defect realisation)**.** *On a Bianchi class-A background ($T^3, \mathrm{Nil}^3, \mathrm{Sol}^3, S^3$), the quadratic constrained perturbation spectrum of the Lorentzian first-order EC+NY system under the DBR protocol satisfies, for each topology:*

(i) Propagating sector. *The physical propagating degrees of freedom are two transverse-traceless (TT) gravitons, and their kinetic block is positive definite.*

(ii) Scalar kernel. *In the scalar (diagonal/axis) sector, the dimension of the non-propagating modes surviving after the Dirac closure is*

$$d_{\mathrm{zero}} = \dim\ker n = 3 - \mathrm{rank}(n), \tag{6}$$

with hierarchy S^3:0 / Sol^3:1 / Nil^3:2 / T^3:3.

(iii) Frame invariance. *d_{zero} is invariant under SO(3) rotations of the spatial frame.*

Scope of the theorem.

- d_{zero} is not claimed as a new topological invariant (the rank defect is known [16]; the content of the theorem is its constraint-dynamical realisation).
- The kernel modes are not identified with propagating physical zero modes. They are a residual non-propagating scalar kernel in the DBR quadratic constrained spectrum (directions unpaired with a kinetic term); their higher-order fate is classified within the finite Palatini branch in Section 7.
- The claim is restricted to class A (no extension to class B).

4.2 Four-topology instances

Topology	Bianchi	rank(n)	d_{zero}	b_1	TT graviton	Zero-mode evidence	TT/propagation evidence ($k > 0$)
T^3	I	0	3	3	2 (pos. kin.)	full-k endpoint	full-k Fourier $\|k\|^2$ (closed form)
Nil^3	II	1	2	2	2 (pos. kin.)	full-spectral scalar (consistent)	closed-form spectral completion over analysed harmonic sectors: toroidal $\cup$ Heisenberg tower $2\pi\|k\|(2N+1)$ (closed form / analytic), with matrix harmonic closure $d(M) = 2$
Sol^3	VI_0	2	1	1	2 (pos. kin.)	full-spectral scalar (positive bulk gap)	bulk gap $2\|pq\|$ (all labels, analytic bound) + compact edge $\lambda_1 > 0$ (representative numerics) + $d(M) = 2$
S^3	IX	3	0	0	2 (pos. kin.)	endpoint (absent for all l)	l-symbolic for all $l \geq 2$ (closed form)

Table 3: Four-topology instances of Theorem 1 (the tier definitions of the evidence columns are in Appendix D).

The "axes" on which the zero modes appear correspond to the kernel directions of the structure matrix: a single singular axis (the zero eigendirection of n) for Sol^3, two axes for Nil^3, all three axes (flat degeneracy) for T^3, and none for S^3. The names of the axes (which coordinate axis) are frame-convention dependent; only the dimension of the kernel is frame-invariant.

4.3 Mechanism: activation rank of the constraint algebra

The mechanism of Theorem 1(ii) starts from writing the scalar-axis perturbation as a vector $\boldsymbol{\beta} \in \mathbb{R}^3$ (components β_a, $a = 1, 2, 3$), with the DBR consistency condition giving a linear map

$$\mathcal{A}_n : \ \boldsymbol{\beta} \longmapsto (\text{secondary activation components}). \tag{7}$$

On the class-A Milnor frame, the formal structure map from the structure constants to the class-A matrix satisfies $\mathcal{A}_{\text{form}} = n$. We do not use this identity alone as the physical DBR activation evidence. Its role as the DBR activation map is checked by the scalar-axis DBR closure audit described in Appendix B.2.

Proof skeleton of Theorem 1(ii).

1. For the scalar-axis sector, the DBR closure is run through primary preservation, secondary generation, the constraint Poisson matrix rank, and the first/second-class split. This determines whether each scalar axis is activated.

2. In the four-topology by three-axis audit, the engine-derived activation matrix has the same rank as n. Hence the number of independent secondary constraints that are activated is $\mathrm{rank}(\mathcal{A}_{\mathrm{DBR}}) = \mathrm{rank}(n)$ (the *activation rank*).

3. The unactivated directions are $\ker n$, of dimension $3 - \mathrm{rank}(n) = d_{\mathrm{zero}}$.

4. Along the $\ker n$ directions no kinetic term paired with the TT kinetic block arises (these coincide with the kernel of the scalar-axis Hessian; Section 6). Hence the surviving modes are not propagating zero modes but a *non-propagating scalar kernel*.

This is not claimed as a fully general symbolic DBR theorem for arbitrary class-A structure constants. For the four class-A backgrounds studied here, the formal identity $\mathcal{A}_{\mathrm{form}} = n$ is reported as a notation and structure map check, while the primary evidence for DBR activation is the axis-by-axis closure output of `script/scripts/paper07/rank_defect_activation_witness.py`.

Remark on terminology. In the constraint analysis of Poincaré gauge theory, "constraint bifurcation" and "field activation" phenomena arising from nonlinear constraints are known [30, 31]. Our activation rank is distinct from these: it is the algebraic statement that the activation dimension of the *linear (quadratic-action) level* secondary constraints coincides exactly with the rank of the structure matrix.

4.4 Frame invariance

Analytic proof. Under an SO(3) rotation R of the spatial frame, the structure matrix undergoes the congruence transformation

$$n \longmapsto R\, n\, R^{\mathsf{T}}. \tag{8}$$

The rank and $\dim\ker n$ are preserved not only by similarity but also by this congruence. Hence $d_{\mathrm{zero}} = 3 - \mathrm{rank}(n)$ is frame-invariant. The basis vectors of $\ker n$ (the axis names) merely rotate under R and are not physical.

Auxiliary numerical check. The above was confirmed by two routes: (a) random SO(3) rotations and (b) reconstruction of n from the rotated structure constants (Appendix B.4). The kernel dimension was preserved in every check.

4.5 Cross-check: $d_{\text{zero}} = b_1$ (not primary)

For class-A algebras, $\dim \ker n = \dim(\mathfrak{g}/[\mathfrak{g},\mathfrak{g}])$, which equals the first Betti number b_1 of the corresponding compact quotient. This isomorphism (Lie-algebra cohomology of left-invariant representatives $\leftrightarrow$ de Rham cohomology of the compact quotient) is due to Nomizu [41] for nilmanifolds and to Hattori [42] for completely solvable solvmanifolds (since $\mathrm{Sol}^3(\mathrm{VI}_0)$ is completely solvable, i.e. of type R, the latter applies and gives $d_{\text{zero}} = b_1 = 1$). This coincidence is an *independent cross-check* of Theorem 1, a geometrically known correspondence. We do not use it as the primary support of the theorem (the primary support is the activation-rank derivation of Section 4.3). The geometric realisation of the kernel is given in Section 6.

4.6 What remains after the quadratic theorem

Theorem 1 is a quadratic-order statement. Two natural questions remain: (a) whether the classification is unchanged for the inhomogeneous modes ($k > 0$), addressed in Section 5; and (b) whether the kernel directions are flat directions (moduli) under finite deformation or are blocked by walls, addressed in Section 7.

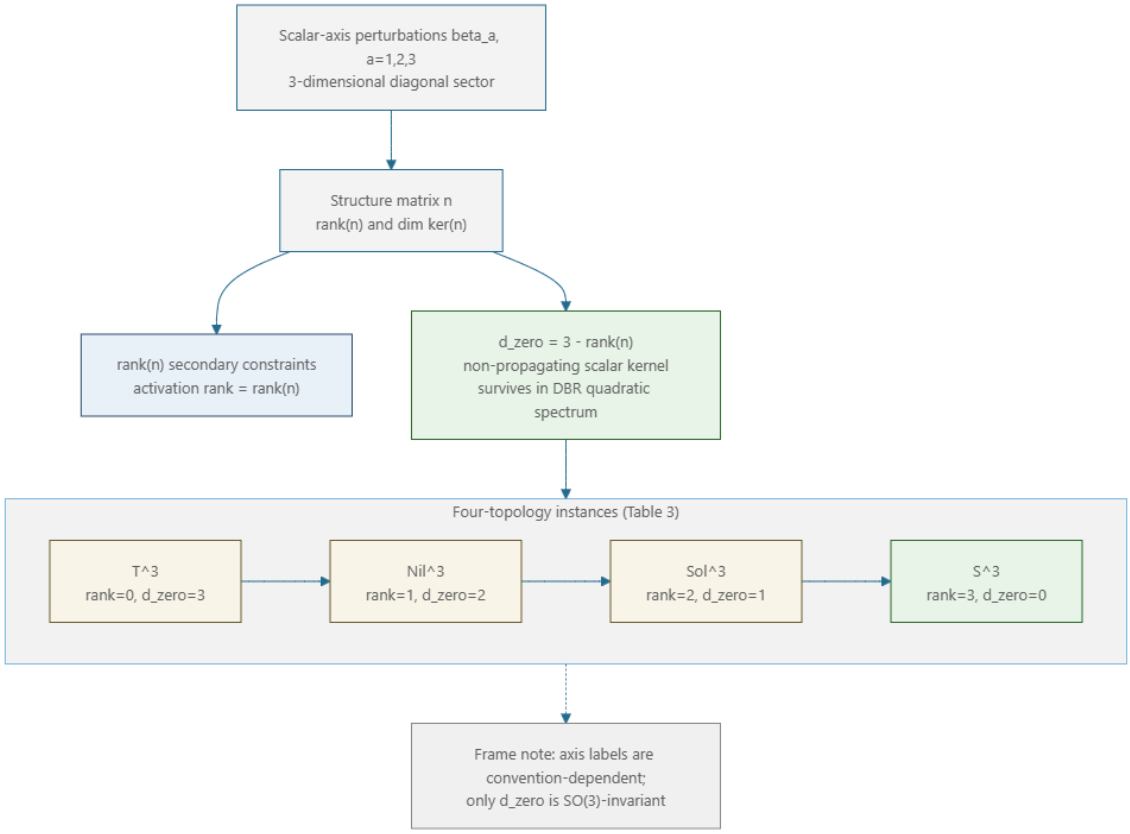


Figure 3: Rank-defect mechanism and four-topology ladder. The consistency condition for the scalar-axis perturbations β_a generates the secondary constraints through the structure matrix n. The $\mathrm{rank}(n)$ image directions are activated (activation rank), and the $d_{\mathrm{zero}} = 3 - \mathrm{rank}(n)$ directions of $\ker n$ survive as the non-propagating scalar kernel. The four topologies form the hierarchy T^3:3 / Nil^3:2 / Sol^3:1 / S^3:0. *Scope*: the axis names are frame-convention dependent; only the kernel dimension is invariant (Section 4.4).

5 Robustness I: scoped $k > 0$ spectral completion

In this section we show that the classification of Theorem 1 is unchanged for the inhomogeneous modes, as a $k > 0$ *spectral completion over the analysed harmonic sectors*. The verification is organised by the following operational definitions:

- **H1 (propagation)**: at the analysed labels the TT propagation operator is positive (for $k > 0$ the two TT modes propagate with positive dispersion).
- **H2 (no-ghost)**: the TT kinetic Hessian is positive definite at all labels.
- **H3 (non-contamination)**: the analysed $k > 0$ sectors introduce no additional normalisable physical zero into $d_{\text{zero}} = 3 - \text{rank}(n)$.

H1–H3 are not a full field-theory spectral theorem; they are a robustness criterion restricted to the analysed harmonic sectors.

5.1 Endpoint controls: T^3 and S^3

The two ends of the four-topology ladder ($d_{\text{zero}} = 3$ and 0) have known spectra, and serve as cross-checks of the method.

- T^3 **(proof-like derivation of H1–H3)**. In the abelian Fourier expansion the TT spatial operator is as below, so for $k \neq 0$ one has $\lambda_{T^3} > 0$ (H1). The kinetic block is positive definite (H2). The $k = 0$ locus is separated as a flat-degeneracy endpoint/gauge locus and is not included in the $k > 0$ claim of H3:

$$\lambda_{T^3}(\boldsymbol{k}) = |\boldsymbol{k}|^2. \tag{9}$$

- S^3 **(similarly)**. By the tensor harmonics [28, 29] the TT eigenvalues are as below, so $\lambda_{S^3} > 0$ for $l \geq 2$ (H1–H2). The scalar zero is absent for all l (l-symbolic; H3). The cases $l = 0, 1$ are classified as endpoint/gauge/non-dynamical (Appendix A):

$$\lambda_{S^3}(l) = (l-1)(l+3) \qquad (C = \epsilon \text{ frame}). \tag{10}$$

5.2 Nil^3: closed-form spectral completion

On Nil^3 (a compact quotient of the Heisenberg group), the analysed harmonic sectors decompose into *two towers*:

- **toroidal sector** ($k = 0$):

$$\lambda_{\text{tor}}(m, n) = 4\pi^2(m^2 + n^2), \qquad (m, n) \neq (0, 0) \;\Rightarrow\; \lambda_{\text{tor}} > 0. \tag{11}$$

- **Heisenberg sector** ($k \neq 0$): by non-commutativity this forms a Landau-type tower, positive ($2\pi|k| > 0$) even at the lowest rung $N = 0$:

$$\lambda_{\text{Heis}}(k, N) = 2\pi|k|\,(2N + 1), \qquad N = 0, 1, 2, \ldots \tag{12}$$

H1–H2 hold in both towers, and the positive gap implies H3 (no additional physical zero). This is consistent with the matrix harmonic closure (TT: $d(M) = 2$, spin-1: $d(M) = 0$; Appendix D). We call this result a "closed-form spectral completion over the analysed harmonic sectors."

5.3 Sol^3: bulk gap and the compact edge

The non-commutative sector of Sol^3 is labelled by boost-type harmonics, and the reduced spatial operator reduces to the one-dimensional Schrödinger form

$$-\frac{\mathrm{d}^2}{\mathrm{d}w^2} + V(w), \qquad V(w) = q^2 e^{2w/R} + p^2 e^{-2w/R}, \tag{13}$$

(p, q are labels). We treat the *bulk* and the *compact edge* separately.

- **Bulk ($pq \neq 0$)**: by the arithmetic–geometric-mean inequality

$$V(w) \geq 2|pq|, \qquad \text{with equality at } |q\,e^{w/R}| = |p\,e^{-w/R}|. \tag{14}$$

  Hence $\inf V = 2|pq| > 0$ (analytic bound, all labels) and H1–H3 hold.

- **Compact edge (representative numerics)**: the physical Sol^3 is a torus bundle with hyperbolic gluing $A \in SL(2, \mathbb{Z})$ (an Anosov mapping torus), and each sector becomes a two-sided confining modified-Mathieu type [40]. For a representative gluing parameter, a finite-difference Sturm–Liouville witness confirms the lowest eigenvalue $\lambda_1 > 0$. This is not an all-Anosov closed-form theorem; it is kept at the level of representative numerics (`script/scripts/paper07/spectral_completion_witness.py`).

For $pq = 0$ there is a gapless continuum with $\inf V = 0$ (non-compact; a signature of amenability [38, 39]); this is not called a normalisable zero and is separate from the compact edge.

5.4 No-ghost across all labels

The TT kinetic Hessian is given, for all topologies, in the form

$$K = \mathrm{diag}\left(\frac{2q^3}{N}\right), \tag{15}$$

label-independent ($\partial K/\partial k = 0$) and positive definite. Hence the no-ghost property extends to all analysed labels. The minisuperspace no-ghost is treated as an independent verification in Section 7.

5.5 Non-contamination of d_{zero}

Since the bulk gap is positive in all analysed harmonic sectors, the $k > 0$ spectral completion introduces no additional physical zero into $d_{\text{zero}} = 3 - \text{rank}(n)$. The homogeneous count of d_{zero} and the full-spectral statement about physical degrees of freedom are propositions at different levels; what this section shows is that the former is unchanged by the latter.

5.6 NY-inhomogeneous note

The $O(\delta^2)$ density of the NY term does not contribute to the rank in the bulk, even in the analysed spectral sectors. This fact is detailed in Section 8.

5.7 Evidence tiers

Tier	Meaning	Instances in this section
Closed form	Closed form / symbolic, all labels	T^3 full-k, S^3 l-symbolic, Nil^3 tower
Analytic bound	Analytic, all labels (incl. non-closed-form bound)	Sol^3 bulk gap $2\lvert pq\rvert$
Representative numerics	Numerical, representative label	Sol^3 compact edge λ_1 (representative)

The full spectral-verification table is collected in Appendix D.

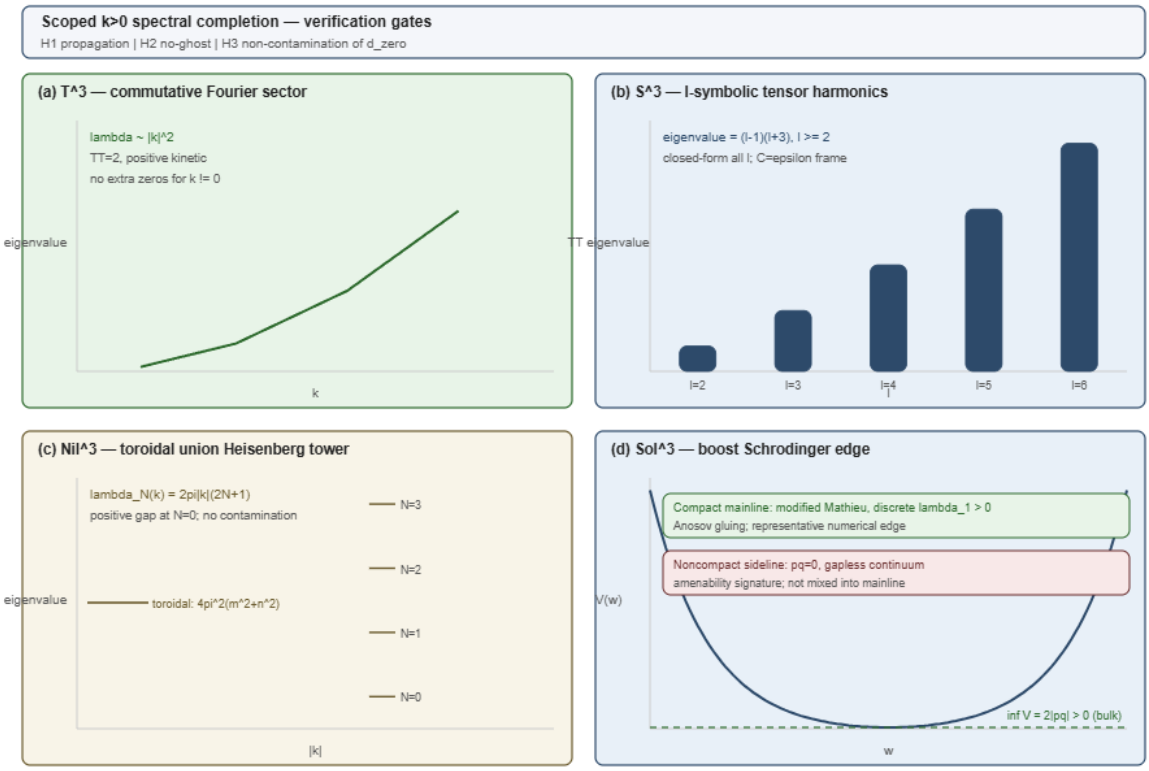


Figure 4: Scoped $k > 0$ spectral completion map. The analysed harmonic sectors of the four topologies and the verification criteria (propagation, no-ghost, non-contamination) are shown. **(a)** T^3: abelian Fourier sector, $|k|^2$ dispersion. **(b)** S^3: l-symbolic tensor harmonics, $(l-1)(l+3)$. **(c)** Nil^3: sum of the toroidal sector and the Heisenberg Landau tower $2\pi|k|(2N+1)$. **(d)** Sol^3: boost Schrödinger potential $V(w)$, with the compact main line (modified Mathieu, discrete $\lambda_1 > 0$) and the non-compact subline ($pq = 0$ gapless continuum) shown separately. *Scope*: this figure is restricted to the analysed harmonic sectors and is not a full PDE theorem for arbitrary inhomogeneous perturbations.

6 Harmonic one-form realisation of the zero kernel

In this section we show how the scalar kernel of Theorem 1 is realised as a geometric object. The claim is within the left-invariant (representative) scope.

6.1 Left-invariant harmonic one-forms and $\ker n$

Lemma 1 (Harmonic one-form $\Leftrightarrow \ker n$)**.** *On the left-invariant coframe $\{\sigma^i\}$ of a Bianchi class-A geometry, the space of left-invariant harmonic one-forms coincides with $\ker n$.*

Proof. For a left-invariant one-form $\xi = \xi_i\,\sigma^i$ (ξ_i constant), the structure equations give

$$\mathrm{d}\xi = -\tfrac{1}{2}\,\xi_i\,C^i{}_{jk}\,\sigma^j \wedge \sigma^k, \tag{16}$$

and since $C^i{}_{jk} = \epsilon_{jkl}n^{li}$ in class A,

$$\mathrm{d}\xi = 0 \quad\Longleftrightarrow\quad n\,\xi = 0 \quad\Longleftrightarrow\quad \xi \in \ker n. \tag{17}$$

Moreover, for a left-invariant metric the co-closed condition $\mathrm{d}{\star}\xi = 0$ is satisfied automatically by unimodularity (class A). Hence the harmonic one-forms are isomorphic to $\ker n$, of dimension d_{zero}. On the compact quotient they provide representatives of the de Rham H^1, realising the geometric content of the coincidence with b_1 (Section 4.5) [41, 42]. □

The agreement of the nullspaces of the exterior-derivative operator d_1 (on left-invariant one-forms) and of n is verifiable with `script/scripts/paper07/harmonic_kernel_correspondence.py`.

6.2 Map to the Hessian kernel

Define the map from the scalar-axis perturbations β_a to the left-invariant one-form directions by

$$\Phi:\ \beta_a \longmapsto \beta_a\,\sigma^a \tag{18}$$

(no sum; Milnor frame).

Proposition 1 (Hessian kernel $\leftrightarrow$ harmonic basis)**.** *The kernel of the scalar-axis Hessian of the reduced quadratic action is in one-to-one correspondence with $\Phi(\ker n)$. The map Φ preserves the subspace (of dimension d_{zero}), not the axis names.*

Proof sketch. For $\beta \in \ker n$, Lemma 1 makes $\Phi(\beta)$ harmonic. By the activation rank of Section 4.3, the β direction does not activate a secondary constraint, and no kinetic pairing arises in the scalar-axis Hessian. Conversely, each direction of the Hessian kernel is realised as a frame stretching along the basis $\xi^{(\alpha)}$ of $\ker n$ (Appendix E). □

6.3 Scope

This realisation is a correspondence at the left-invariant (homogeneous representative) level, not a full Hodge-theoretic statement including the inhomogeneous modes. That no additional zero arises in the inhomogeneous sectors is independently guaranteed by the non-contamination criterion of Section 5.

7 Finite fate of the zero kernel

After finding d_{zero} non-propagating kernel directions at quadratic order, the natural question is whether these are flat directions (moduli) under finite deformation, or are blocked by walls at finite amplitude. This section answers that question *within the finite Palatini branch / homogeneous minisuperspace scope. This section is a supporting result and is not part of Theorem 1 (the quadratic theorem).*

7.1 Classification: exact moduli versus curvature walls

For a finite deformation $\beta \mapsto$ finite value along a kernel direction, we reconstruct a background-adapted static Palatini branch and verify the finite expansion (symbolic, up to fourth order) of the reduced energy density $\Delta(\beta)$. The result is the following dichotomy:

- **Exact modulus**: $\Delta(\beta) = 0$ holds exactly at all orders of the expansion (a flat direction persisting under finite deformation).
- **Curvature wall**: $\Delta(\beta)$ develops a curvature-induced non-zero term (including a tadpole), and no stationary finite-deformation branch exists.

Key points at the level of formulas (finite Palatini branch / homogeneous minisuperspace).

- T^3: the spatial scalar curvature vanishes identically. Hence $\Delta(\beta) \equiv 0$ for any kernel direction (exact flat homogeneous branch).
- Nil^3 β_-: along the tangent vector $(\frac{1}{2}, -\frac{1}{2}, 0)$ in the Milnor frame, $\Delta(\beta_-) = 0$ (residual 0 up to fourth order; exact modulus). The closed-form branch formula is in Appendix F.3.
- Nil^3 β_+/u: along $(-\frac{1}{2}, -\frac{1}{2}, 1)$ a curvature-induced linear term (tadpole) remains in the reduced energy; a curvature wall.
- Sol^3 β_x: along $(\frac{1}{3}, -\frac{1}{6}, -\frac{1}{6})$ a non-zero term remains; no stationary branch (curvature wall).

Topology	d_{zero}	Exact moduli	Curvature wall	Remarks
T^3	3	3	0	exact flat homogeneous moduli
Nil^3	2	1 (β_-)	1 (β_+/u)	β_- exact modulus (residual 0 to fourth order); β_+ wall with tadpole
Sol^3	1	0	1 (β_x)	stationary branch absent
S^3	0	0	0	$d_{\text{zero}} = 0$ endpoint (not applicable)

Table 4: Modulus/wall split (finite Palatini branch / homogeneous minisuperspace scoped).

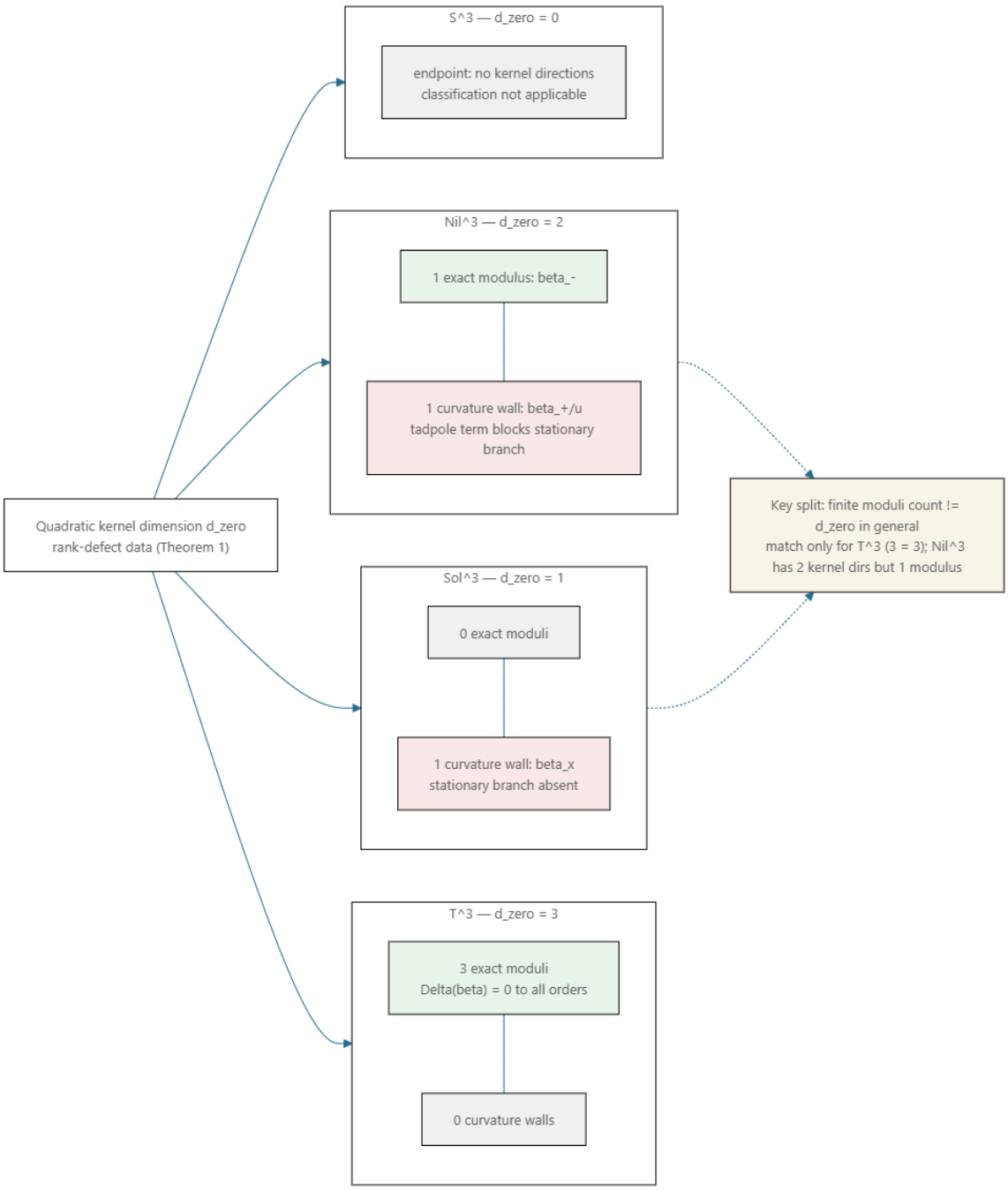


Figure 5: Modulus/wall split across the four topologies. The correspondence between the quadratic kernel dimension d_{zero} and the exact moduli / curvature walls on the finite Palatini branch is shown. d_{zero} is preserved as rank-defect data and does not in general equal the finite moduli count (they agree only for T^3). *Scope*: the wall checks are restricted to the finite Palatini branch / homogeneous minisuperspace; this is not a full nonlinear lift theorem.

7.2 Minisuperspace DBR closure

The consistency of this classification was verified with the DBR closure of a lever-complete homogeneous minisuperspace ansatz including the lapse, the Hamiltonian constraint, and

the DeWitt metric. The results: (i) the closure including the Hamiltonian constraint holds; (ii) the DeWitt norm of the trace-free tangent directions is positive (minisuperspace no-ghost); and (iii) the configuration degrees of freedom of the homogeneous anisotropy are 2 (verification data in Appendix F).

7.3 Mandatory caveats

- **The finite moduli count is not in general equal to d_{zero}.** They agree only for T^3 (3=3); for Nil^3 the exact modulus is 1 against $d_{\text{zero}} = 2$. d_{zero} is preserved as rank-defect data (the dimension of the quadratic kernel) and is a different quantity from the finite modulus count.
- **The wall checks are established within the finite Palatini branch**, and are not a full nonlinear lift theorem or mass theorem. No extension to the time-dependent full DBR dynamics or to the inhomogeneous sector is made.
- **The minisuperspace no-ghost is not a full field-theory no-ghost** (the analysed scope of the latter is Section 5).
- The energy-density-expansion classification and the stationarity decision in the branch reconstruction are distinct checks, presented separately in Appendix F.

7.4 Torsion non-contamination (corollary)

The torsion modes (EH/AX/VT/MX) do not change this modulus/wall split. This is a corollary of the triviality of the torsion auxiliary shell (not an independent dynamical check); its precise statement is in Section 8.

8 Torsion/Nieh–Yan sector robustness

Although the action contains the NY term and torsion, the theorem and robustness checks so far have not referred to torsion. In this section we examine the effect of the torsion sector on the preceding results. The claim is confined to a single point: *the torsion sector affects none of the theorem hierarchy of this paper (quadratic / $k > 0$ spectral completion / finite deformation).* The physics of torsion itself (boundary charge, localisation, etc.) is outside the scope of this paper.

8.1 Known/new boundary

That the NY term is topological in the canonical analysis and does not change the classical dynamics is known [33–35]. The content of this section is not a restatement of that generality but an *exact demonstration in the reduced system*: (i) the reduced action is exactly at most quadratic in the torsion ansatz scalars; (ii) under field promotion no derivative coupling arises in the inspected criterion; and (iii) the finite-deformation classification is invariant.

8.2 (i) Exact auxiliary shell

The reduced torsion Lagrangian of the NY-MAIN route can be organised as an auxiliary shell in the form

$$\mathcal{L}_{\text{aux}} = a_\eta\,\eta^2 + a_V\,V^2 + b_V\,\dot{\text{vol}}\,V + (\text{boundary total-derivative}), \tag{19}$$

(the coefficients a_η, a_V, b_V are topology/mode dependent but algebraic in η, V). By a polynomial-degree verification (symbolic up to degree 4, residual zero), it is *exactly at most quadratic* in (η, V).

Algebraic proof (shell condition). The Euler–Lagrange equations are

$$\frac{\partial\mathcal{L}_{\text{aux}}}{\partial\eta} = 2a_\eta\,\eta = 0, \qquad \frac{\partial\mathcal{L}_{\text{aux}}}{\partial V} = 2a_V\,V + b_V\,\dot{\text{vol}} = 0. \tag{20}$$

If $a_\eta \neq 0$ then $\eta^* = 0$. The linear term $b_V\,\dot{\text{vol}}\,V$ in V can be separated as the boundary current

$$2\,\frac{\mathrm{d}}{\mathrm{d}t}(\text{vol}\cdot V)\,, \tag{21}$$

and the bulk equation gives $V^* = 0$. Hence

$$\eta^* = 0, \qquad V^* = 0 \tag{22}$$

is an *exact auxiliary shell*, not an approximation (Appendix G).

8.3 (ii) Field promotion does not generate kinetic terms

Decompose the connection as $\Gamma = \Gamma_{\mathrm{LC}} + K(\eta, V)$, with

$$R[\Gamma] = \mathrm{d}\Gamma + \Gamma \wedge \Gamma, \qquad K = K(\eta, V) \text{ (algebraic in the AX/VT/MX ansatz scalars).} \tag{23}$$

If K is algebraic in η, V then dK is first order in the field derivatives, and the quadratic derivative couplings $(\dot{\eta}^2, \dot{V}^2, (\nabla\eta)^2, (\nabla V)^2)$ in the scalar contraction of ΔR cancel structurally (a native-action-gate statement; not a full torsion field theory).

Promoting the scalars to fields $\eta(x), V(x)$, we inspected the above over all cells of four topologies $\times$ AX/VT/MX. The results:

- No quadratic derivative coupling appears at all. Even a linear spatial gradient is absent.
- What appears is only a boundary-type time-linear current: $2\,\mathrm{vol}\cdot\dot{V}$ on the EC side and $6\,\mathrm{vol}^{2/3}\,\dot{\eta}$ on the NY side.
- At the level of the Riemann components individual derivative terms fire, but they cancel under the scalar contraction, as confirmed by a component computation.
- The classification procedure is validated by positive-detector controls in the native-action gate that the witness checks before reporting the cells above: a free-scalar toy run through the same classifier recovers the expected positive time-kinetic and negative spatial-gradient coefficients (alongside ghost, wrong-sign, and linear-current toys), so a genuine kinetic term would be detected (`script/scripts/paper07/torsion_auxiliary_field_promotion_witness.py`; no claim about a full 24-component torsion field theory).

8.4 (iii) Finite fate is unchanged

The modulus/wall split of Section 7 is invariant under all of the torsion modes EH/AX/VT/MX (a corollary of $\mathcal{L}_{\mathrm{aux}}|_{\mathrm{shell}} = 0$ on the auxiliary shell).

8.5 Conclusion and scope

Within the NY-MAIN route and the AX/VT/MX torsion ansatz scalars, the reduced torsion sector remains an exact auxiliary, and its promotion to fields $\eta(x), V(x)$ generates no quadratic kinetic/gradient term in the inspected native-action criterion. Hence torsion/NY affects none of the DBR rank-defect kernel

(Section 4), the robustness from the $k > 0$ spectral completion (Section 5), or the finite homogeneous modulus/wall split (Section 7).

Scope (explicit). The subject of this section is the AX/VT/MX ansatz scalars, not a $(\partial T)^2$-absence theorem for a full 24-component torsion field theory. The exact-degree-2 / exact-shell claim is restricted to the NY-MAIN route. A time-linear boundary current is not called a kinetic/propagating mode. The absence of NY bulk involvement is not claimed outside this scope.

Layer	Check	Result	Scope
Quadratic	Polynomial-degree verification (η, V)	exact degree-2; $\eta^* = V^* = 0$ exact shell	NY-MAIN route
Finite	Invariance of the modulus/wall split	invariant (corollary)	finite Palatini branch
Field-promoted	Derivative-coupling classification (4 topo. $\times$ AX/VT/MX)	kinetic/gradient absent; boundary current only	inspected native-action criterion

Table 5: Hierarchy of the torsion robustness.

9 Discussion and outlook

9.1 What has been established, by layer

The results of this paper can be organised along their logical layers of establishment. All of the following are claims under the DBR protocol.

Quadratic level (Sections 4 and 6). In the DBR quadratic constrained spectrum, we proved that the dimension of the non-propagating scalar kernel is realised as $d_{\text{zero}} = \dim\ker n = 3 - \text{rank}(n)$. We also showed that two TT modes propagate with positive kinetic, that d_{zero} is invariant under SO(3) rotations of the spatial frame, and the activation-rank mechanism. At this layer we do not identify the kernel with propagating zero modes, do not claim d_{zero} as a new invariant, and do not extend to class B.

$k > 0$ spectral completion (Section 5). In the analysed harmonic sectors we confirmed the propagation of the 2 TT, the no-ghost property, and the non-contamination of d_{zero}, through the endpoint controls (T^3, S^3), the closed-form completion of Nil^3, and the bulk gap and positive compact edge of Sol^3. This is a robustness restricted to the analysed harmonic sectors (the Sol^3 edge is representative numerics), and is not a claim of full PDE closure, an all-Anosov closed-form edge theorem, or a compact continuous spectrum.

Classification of finite deformations (Section 7). Within the finite Palatini branch / homogeneous minisuperspace we exhibited the modulus/wall split (T^3 3/0, Nil^3 1/1, Sol^3 0/1, S^3 0/0) and the minisuperspace no-ghost. This is a supporting result and does not include a nonlinear lift/mass theorem or an identification of the moduli count with d_{zero}.

Torsion closure (Section 8). Within the inspected AX/VT/MX ansatz, we showed that the reduced torsion sector remains an exact auxiliary shell, generates no quadratic kinetic term under field promotion, and leaves the finite-deformation classification invariant. This is not a full 24-component $(\partial T)^2$-absence theorem, a general claim of NY bulk non-involvement, or a charge/localisation theorem.

Separately, the route comparison (Section 3, Appendix C) shows, as a control, that the RBD route produces a Dirac-inequivalent extra mode in the S^3 scalar sector. This is the basis for adopting DBR and does not claim a general failure of RBD or a physical-ghost interpretation.

9.2 Scope of the novelty (restated)

The topology hierarchy S^3:0 / Sol^3:1 / Nil^3:2 / T^3:3 of this paper is, algebraically, the *known data* of the rank-defect / abelianisation / b_1 of the Bianchi class-A structure

matrix [16, 41, 42]. Our contribution is not its discovery but the three points: (i) the verification that, in the Lorentzian first-order EC+NY system, the DBR constraint dynamics *independently reproduces* this data as a residual non-propagating scalar kernel; (ii) the four-topology comparison under a single protocol; and (iii) that this classification is protected by the three-way robustness checks of the $k > 0$ spectral completion, the finite-deformation classification, and the torsion closure.

9.3 Relation to Teichmüller/moduli counts of compact Bianchi models

The global degrees of freedom (Teichmüller parameters) of compact Bianchi models have been systematically counted by Koike–Tanimoto–Hosoya [24–26]. Our d_{zero} is a *different object*: the KTH moduli are global structural deformations of the compact quotient of metric GR, whereas our kernel is the non-propagating direction of the homogeneous perturbations in the DBR constrained spectrum of the first-order EC+NY system. Indeed, internally to this paper, the finite exact moduli count (T^3:3, Nil^3:1, Sol^3:0) does not in general equal d_{zero} (3, 2, 1) (Section 7). The precise correspondence among the quadratic kernel, the finite moduli, and the Teichmüller deformations (which kernel directions lift to global deformations) remains a task for future work.

9.4 Outlook

- **Higher-order lift**: a full nonlinear lift/mass analysis beyond the finite Palatini branch (to settle the dynamical meaning of the wall).
- **Extension to class B / VII_0 / VIII**: our formula is restricted to class A. In class B there are known obstructions to the symmetry reduction itself [21], requiring a separate treatment. It is also known that the Mathieu structure appears in the spectral problem of the Bianchi VII system.
- **Closed form for the Sol^3 edge**: an upgrade of the compact edge λ_1, kept at representative numerics, to an all-Anosov closed-form theorem.
- **Correspondence with the Euclidean series**: a comparison of the results of this paper (Theorem 1, Table 3, Table 4, Table 5) with the Euclidean response dictionary [4, 5].

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The AI tools did not determine the scientific claims of this work and were not used to generate or modify research data or evidentiary figures. The author takes full responsibility for the content and for any remaining errors.

This work was developed within the informal collaborative project **DPPU** (**D**onut-like topology, **P**lanck-scale compactness, **P**recession dynamics, **U**niverse).

Availability and Contact

Code and Data Availability The code and supplemental data associated with this work are available at:

- Zenodo: `10.5281/zenodo.20678646`
- GitHub: `https://github.com/Muacca/DPPUv7-paper07`

They are released under the MIT License.

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> Muacca, "Dirac-before-Routh Constrained Spectra in Bianchi Class-A Einstein–Cartan–Nieh–Yan Cosmologies: Rank-Defect Kernels, Spectral Robustness, and Torsion-Sector Closure"

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- GitHub Issues: `https://github.com/Muacca/DPPUv7-paper07/issues`

A Conventions

A.1 Signature, Hodge, and ϵ conventions

- Metric signature: $(-,+,+,+)$ (Lorentzian).
- Internal indices $I, J, \dots \in \{0,1,2,3\}$; spatial frame indices $a, b, \dots \in \{1,2,3\}$.
- Levi-Civita symbol: $\epsilon_{0123} = +1$; in three spatial dimensions $\epsilon_{123} = +1$.
- The Hodge dual is defined in the Lorentzian convention, with the four-form orientation $e^0 \wedge e^1 \wedge e^2 \wedge e^3$ taken positive.
- Curvature/torsion: $R^{IJ} = \mathrm{d}\omega^{IJ} + \omega^I{}_K \wedge \omega^{KJ}$, $T^I = \mathrm{d}e^I + \omega^I{}_J \wedge e^J$.

A.2 Structure matrix sign convention

For the structure equation $\mathrm{d}\sigma^i = -\frac{1}{2} C^i{}_{jk} \sigma^j \wedge \sigma^k$, we define

$$n^{ij} = \tfrac{1}{2}\, \epsilon^{ikl}\, C^j{}_{kl} \tag{24}$$

(symmetric in class A). The signs of the diagonal values in the Milnor frame are fixed to the convention of Table 1 (Nil^3: $\mathrm{diag}(1,0,0)$, Sol^3: $\mathrm{diag}(1,-1,0)$, S^3: $\mathrm{diag}(1,1,1)$). $\mathrm{rank}(n)$ and $\dim\ker n$ do not depend on the normalisation (scale) of the diagonal values.

A.3 S^3 normalisation: the $C = \epsilon$ frame

The S^3 structure constants are fixed throughout the paper to

$$C^i{}_{jk} = \epsilon^i{}_{jk} \qquad (|C| = 1;\ C = \epsilon \text{ frame}). \tag{25}$$

Scaling and invariants. Under an overall scaling of the structure constants $C \to sC$ ($s > 0$) one has $n \to sn$, but

$$\mathrm{rank}(n), \qquad \dim\ker n, \qquad d_{\mathrm{zero}} = 3 - \mathrm{rank}(n) \tag{26}$$

are invariant. By contrast, the *numerical* values of the curvature and the eigenvalues are scale-dependent. Representative formulas:

$$R\, q^2 \ \mapsto\ s^2\, (R\, q^2) \qquad (\text{isotropic point}), \tag{27}$$

$$\lambda_{S^3}(l) = (l-1)(l+3) \ \mapsto\ s^2\, (l-1)(l+3) \qquad (\text{TT sector};\ l \geq 2). \tag{28}$$

Every place where we write $R\,q^2 = 6$ and TT eigenvalue $(l-1)(l+3)$ refers to the value in the $C = \epsilon$ ($s = 1$) frame. The script `script/scripts/paper07/s3_normalization_factor_witness.py` independently confirms the rank invariance and the s^2 scaling of the curvature and eigenvalues.

Correspondence with paper VI. The S^3 numerics of paper VI [7] (e.g. $\chi = 4$, $C_{\text{topology}} = -36$, $V_3 = 2\pi^2$) are based on a different structure-constant normalisation. The two are related by a scaling of the structure constants, with an explicit factor-16 conversion entering the second-order curvature quantities. We do not quote the S^3 values of paper VI without the explicit conversion; the factor conversion is confined to this appendix.

A.4 Harmonic conventions

- T^3: Fourier modes $e^{ik\cdot x}$, $k \in 2\pi\mathbb{Z}^3/L$.
- S^3: the scalar/vector/tensor harmonics follow the convention of [28, 29]; the TT tensor-harmonic eigenvalues are written as $(l-1)(l+3)$ $(l \geq 2)$ in the $C = \epsilon$ frame.
- $\text{Nil}^3/\text{Sol}^3$: the toroidal $\cup$ Heisenberg sector and the boost sector, respectively, are collected in Appendix D.

A.5 Diagnostic exclusions

This paper does not introduce Pontryagin-type diagnostics (P_{form}, P_{int}, etc. [5–7]). A minimal definition of the NY boundary three-form appears only in Appendix G (for precision).

B Native local action and DBR computational checks

B.1 Reduced action ingredients

Under the homogeneous background ansatz (Section 2.3), the first-order EC+NY action reduces to a finite-dimensional reduced Lagrangian. The independent variables are the scale factors $q_a(t)$, the lapse $N(t)$, the homogeneous connection components, and the torsion ansatz scalars (η, V). The perturbation analysis uses the quadratic action obtained by block-decomposing this reduced system into scalar / spin-1 / spin-2 sectors.

B.2 Symbolic Dirac–Bergmann engine

The constraint analysis was performed with a symbolic computation engine. The procedure follows the standard Dirac–Bergmann algorithm [10–12]:

1. Legendre transform and extraction of the primary constraints.
2. Generation of the secondary constraints by the consistency conditions (iterated to closure).
3. Classification of first/second class, and construction of the Dirac bracket for the second-class pairs.
4. Presentation of the reduced spectrum by Routh-type elimination after closure (the DBR order).

The decisions of the engine do not fix in advance the expected values of any particular topology. In the implementation, each topology local builder is passed through the scalar-axis DBR closure, and the activation entry is read from primary preservation, secondary generation, the constraint Poisson matrix rank, and the first/second-class split. The indeterminate-symbol relation between the structure constants and the class-A matrix is checked separately as a formal identity. The physical role of that map as DBR activation is checked by the axis-by-axis closure audit.

Correspondence with the proof skeleton of Section 4.3. Write $\mathcal{A}_{\mathrm{DBR}}$ for the secondary activation map for the scalar-axis perturbations $\boldsymbol{\beta} \in \mathbb{R}^3$. In the class-A Milnor frame, the formal structure-map identity gives

$$\mathcal{A}_{\mathrm{form}} = n. \tag{29}$$

This identity is a notation and structure-map check; it is not used as a substitute for DBR closure. The witness `rank_defect_activation_witness.py` calls the bundled

DBR constraint-closure engine (`dppu.dirac`) and computes the entries of $\mathcal{A}_{\mathrm{DBR}}$ from DBR closure for the four topologies and three scalar axes. The resulting engine-derived activation matrix satisfies $\mathrm{rank}(\mathcal{A}_{\mathrm{DBR}}) = \mathrm{rank}(n)$ and $\dim\ker(\mathcal{A}_{\mathrm{DBR}}) = d_{\mathrm{zero}}$.

B.3 Rank verification conventions

For the rank decisions of the constraint matrices and the kinetic Hessians we impose the following redundancies:

- **Two routes**: agreement of the symbolic rank and the numerical rank is required.
- **Two generic points**: the numerical rank is evaluated at two generic background points, and agreement is required (excluding an accidental single-point degeneracy).
- **Mod-p check**: for large symbolic matrices, a rank computation at a prime modulus is used in parallel to exclude a spurious rank from cancellation/round-off.

B.4 Frame-invariance check protocol

The frame invariance of Section 4.4 is based first on the analytic argument that $\mathrm{rank}(n)$ and $\dim\ker n$ are preserved under the congruence $n \mapsto RnR^{\mathsf{T}}$. The numerical verification reinforces it by two routes: (a) random SO(3) rotations ($\dim\ker n$ preserved in all 20 rotations), and (b) reconstruction of n from the rotated structure constants (tensor transformation) and recomputation of the kernel dimension. The script `rank_defect_activation_witness.py` also confirms $\mathrm{rank}(RnR^{\mathsf{T}}) = \mathrm{rank}(n)$ for rational orthogonal matrices R.

B.5 Comparison procedure under a single protocol

The comparison of the four topologies was carried out under one and the same engine, the same sector decomposition, and the same rank-verification conventions. S^3 doubles as a cross-check of the method, passing the null check of the scalar-axis projection (all axes zero for the isotropic S^3).

B.6 Reproducibility notes

The main formulas and decisions in the paper are reproducible by the independent witness scripts under `script/scripts/paper07/`.

Witness	Corresponding sections
`rank_defect_activation_witness.py`	§4.3–4.4, B.2–B.4; *1
`spectral_completion_witness.py`	§5, D
`torsion_auxiliary_field_promotion_witness.py`	§8, G
`finite_fate_branch_witness.py`	§7, F
`harmonic_kernel_correspondence.py`	§6, E
`route_comparison_s3_branch_a_witness.py`	§3.3, C
`s3_normalization_factor_witness.py`	§2.2, A.3

Table B.1: Independent witness scripts and the sections they reproduce. (*1: formal identity as notation check, DBR closure audit as primary evidence)

C RBD route comparison and route-dependent discrepancy

This appendix gives the details of the control experiment of Section 3. Here "Palatini reduction non-commutativity" is a route-level descriptive term, not a claim of a general theorem.

C.1 The RBD route

The RBD (Routh-before-Dirac) route solves the auxiliary components of the spatial connection (spatial-ω) by their field equations and substitutes them into the action, and then performs the constraint analysis with the remaining variables. This substitution has not passed the verification of the auxiliary-variable condition (algebraic solvability free of derivatives), and so incurs the non-commutativity risk warned of by the generality [13].

C.2 Direct route recomputation

Our negative control is not a toy model. It uses the bundled DBR/RBD constraint-dynamics engine (`dppu.dirac`) and recomputes the route variants from the same S^3 Branch-A local first-order Lagrangian. The reproducibility script `script/scripts/paper07/route_comparison_s3_branch_a_witness.py` does not use intermediate CSV files as evidence. It directly executes:

- **DBR_KEEP_ALL**: keep the independent connection through Dirac closure;
- **RBD_SPATIAL_OMEGA_ROUTH**: Routh-eliminate spatial ω first, then run scalar projection and Dirac closure;
- **RBD_FULL_K_ROUTH**: Routh-eliminate full K first, then run scalar projection and Dirac closure.

Route	N_{phys} at $l = 0/1/\geq 2$	$l = 0$ pair	H_{PP} at $l = 0$	Status
`DBR_KEEP_ALL`	0/0/0	none	n/a	computed
`RBD_SPATIAL_OMEGA_ROUTH`	1/1/2	$(s_{l0}, P_{s,l0})$	$-3/8$	computed
`RBD_FULL_K_ROUTH`	1/1/2	$(s_{l0}, P_{s,l0})$	$-3/8$	computed

Table C.1: Direct route recomputation in the S^3 Branch-A scalar sector.

The primary evidence for route non-commutativity is the mismatch in N_{phys}. The $l = 0$ value $H_{PP} = -3/8$ is a diagnostic that the RBD extra scalar branch is not a benign rewriting. Since RBD is not the adopted route, this negative sign is not used as a physical ghost or physical instability claim.

C.3 Comparison setup and result

The control experiment was carried out on the S^3 background, Branch-A scalar sector. The direct recomputation of C.2 reproduces the order-of-operations dependence in the DBR/RBD engine.

- **DBR side**: after the Dirac closure, the physical scalar degrees of freedom are 0 at all harmonic levels (the kernel is absent for S^3; $d_{\text{zero}} = 0$).
- **RBD side**: spatial-ω Routh and full-K Routh both give $N_{\text{phys}} = 1/1/2$, so they do not commute with DBR.
- **Same-Lagrangian internal demonstration**: at $l = 0$ the comparison starts from the same Branch-A local first-order Lagrangian and reproduces DBR:0 / RBD:1 together with $H_{PP} = -3/8$ (by symbolic computation, with no numerical error). This establishes that the discrepancy arises from the route itself, not from an implementation or normalisation difference.
- **Scope restriction**: for $l \geq 1$ the RBD extra count is recorded as a route-level artefact involving Lorentz-gauge breaking by the elimination-first frame. The negative-kinetic diagnostic is claimed only at $l = 0$.

C.4 Route comparison labels

Label	Content	Treatment
RBD scalar discrepancy	RBD's extra scalar mode ($H_{PP} = -3/8$ at $l = 0$)	not a physical instability (route-level); not used for physical claims
RBD higher-level artefact	extra RBD counts for $l \geq 1$	recorded as elimination-first gauge breaking; no sign claim promoted
DBR standard	standard route with Dirac closure first	the basis of all claims of this paper

Table C.2: Route comparison labels.

C.5 Scope and limitations (route sector)

- We do not use the output of the RBD route as the basis of any physical claim, positive or negative. We therefore do not treat the RBD extra mode as a physical ghost / physical instability.
- We do not claim the generality that "RBD always fails." It is treated only as a control experiment on a verified sector.

D Endpoint and spectral evidence

This appendix collects the spectral evidence of Sections 4–5. The terminology is unified as the "$k > 0$ spectral completion over the analysed harmonic sectors." The verification criteria H1/H2/H3 correspond to the operational definitions at the start of Section 5.

D.1 Endpoint evidence

- T^3 **(full-k)**: in the Fourier sector the TT spatial operator $= |k|^2$ is confirmed symbolically. The $k = 0$ locus is separated (flat degeneracy; the all-three-axis scalar-axis zero modes are a homogeneous-level result).
- S^3 **(l-symbolic)**: the absence of the scalar-sector zero mode is confirmed symbolically for all l (not a finite sample); the spin-2 TT has eigenvalue $(l-1)(l+3)$ for all $l \geq 2$ with positive definite kinetic Hessian. The cases $l = 0, 1$ are endpoint classifications (gauge/non-dynamical).

D.2 Scalar full-spectral evidence ($\mathrm{Nil}^3/\mathrm{Sol}^3$)

For the scalar modes of the non-commutative harmonic sectors, the effective zero-mode count $d(M)$ is evaluated by the matrix harmonic closure:

- Nil^3: all non-commutative modes have a positive gap ($d(M) = 0$) $\Rightarrow$ the homogeneous-representative kernel count ($\dim\ker n = 2$) agrees with the full-spectral count (consistent).
- Sol^3: the bulk has a positive gap ($d(M) = 0$); the lowest-level truncation-edge term is not counted as a theorem zero (scope-restricted).

D.3 TT / spin-1 full-spectral evidence ($\mathrm{Nil}^3/\mathrm{Sol}^3$)

Extending the same matrix harmonic closure to the spin-2 / spin-1 sectors:

- TT: $d(M) = 2$ (both topologies) – full-spectral consistency of the propagating 2 TT.
- spin-1: $d(M) = 0$ (both topologies) – no physical degrees of freedom in the transverse vector.

D.4 Spectral towers (breakdown of the $k > 0$ completion)

Topology	Sector	Operator / spectrum	Grade
T^3	Fourier $k \neq 0$	$\lvert k\rvert^2$	closed form (endpoint control)
S^3	$l \geq 2$ TT	$(l-1)(l+3)$	closed form (endpoint control)
Nil^3	toroidal ($k = 0$)	$4\pi^2(m^2+n^2)$	closed form
Nil^3	Heisenberg ($k \neq 0$)	$2\pi\lvert k\rvert(2N+1)$	closed form / analytic
Sol^3	boost bulk ($pq \neq 0$)	$\inf V = 2\lvert pq\rvert > 0$	analytic bound (all labels)
Sol^3	compact edge	two-sided confining modified Mathieu; discrete, $\lambda_1 > 0$ [40]	closed form / analytic (discreteness, positivity) + representative numerics (explicit λ_1, representative A)
Sol^3	non-compact subline ($pq = 0$)	gapless continuum = signature of amenability [38, 39]	interpretive line (separated from the main line; not counted in the theorem list)

Table D.1: Spectral towers across the four topologies.

D.5 Spectral verification summary

Criterion	T^3	S^3	Nil^3	Sol^3
Propagation	closed form	closed form	closed form / analytic	analytic bound (bulk) + representative numerics (edge)
No-ghost	closed form	closed form	closed form / analytic	analytic bound (bulk; edge TT sign positive)
Non-contamination	closed form	closed form	closed form / analytic	analytic bound (bulk; edge carrier separated)

Table D.2: Spectral verification summary.

D.6 Independent spectral witness

`script/scripts/paper07/spectral_completion_witness.py` reads no external spectral table; it constructs the formulas of Section 5 in-script (T^3: $|k|^2$, S^3: $(l-1)(l+3)$, Nil^3 two towers, Sol^3 bulk gap $\inf V = 2|pq|$ and the representative finite-difference edge) and decides the H1–H3 positivity conditions.

E Harmonic form and Hessian kernel realisation

This appendix collects the verification tables for the correspondence of Section 6.

E.1 Lemma 1 and Proposition 1

For Lemma 1 (nullspace$(d_1) = \ker n$) and Proposition 1 (the map $\Phi : \beta_a \mapsto \beta_a \sigma^a$ relating the Hessian kernel and the harmonic basis), we confirmed the following for each topology:

1. **Closed condition**: for a left-invariant one-form $\xi = \xi_i \sigma^i$, $\mathrm{d}\xi = 0 \Leftrightarrow n\xi = 0$, with agreement of the exterior-derivative matrix d_1 and n, nullspace(d_1) = nullspace(n).
2. **Co-closed condition**: under class A (unimodular), a left-invariant one-form is automatically co-closed ($\mathrm{d} \star \xi = 0$).
3. **Hessian kernel dimension**: the kernel dimension of the scalar-axis Hessian = $\dim \ker n$ (a subspace decision, not an axis-name decision).
4. **Correspondence of Φ**: one-to-one (up to a change of basis) between the basis of $\ker n$ and the Hessian kernel directions.
5. **Dimension agreement**: $\dim\{\text{harmonic one-forms}\} = d_{\text{zero}} = b_1$ (T^3:3, Nil^3:2, Sol^3:1, S^3:0).

Reproduction: `script/scripts/paper07/harmonic_kernel_correspondence.py`.

E.2 Per-topology kernel directions (Milnor frame)

Topology	n	Basis of $\ker n$	Harmonic one-forms
T^3	diag$(0,0,0)$	e_1, e_2, e_3	$\sigma^1, \sigma^2, \sigma^3$
Nil^3	diag$(1,0,0)$	e_2, e_3	σ^2, σ^3 (the fibre direction σ^1 is non-harmonic)
Sol^3	diag$(1,-1,0)$	e_3	σ^3 (single singular axis)
S^3	diag$(1,1,1)$	$\{0\}$	none

Table E.1: Per-topology kernel directions and harmonic one-forms (Milnor frame).

The axis names depend on the Milnor frame convention; only the dimension is frame-invariant (Section 4.4).

E.3 Scope note

This correspondence is a realisation at the left-invariant representative level. The de Rham H^1 representativeness on the compact quotient relies on standard theory [18], and that no additional zero arises in the inhomogeneous sectors is independently guaranteed by the non-contamination criterion of Section 5.

F Finite-fate branch data

This appendix collects the finite-deformation branch data of Section 7. The scope is the finite Palatini branch / homogeneous minisuperspace.

F.1 Branch parametrisation and tangent alignment

For each kernel direction we reconstruct a background-adapted static Palatini branch with the finite-deformation parameter β (and an auxiliary direction u where needed). In each case we confirmed that the tangent vector of the branch is aligned with the kernel direction of the quadratic analysis (tangent alignment).

F.2 Energy-density expansion versus stationarity

- **Energy-density expansion**: evaluation of the reduced energy density $\Delta(\beta)$ after solving the auxiliary-field equations (auxiliary shell). This is the main verification of this appendix.
- **Stationarity decision**: a decision of whether the finite-deformation point is a stationary point of the field equations.

These are distinct checks; we do not reread the result of the energy-density expansion (the value of Δ) as a theorem of the stationarity decision (the existence of a stationary branch).

F.3 Branch classification summary

Label	Tangent vector $\boldsymbol{\beta}$	Classification
T^3 e_a	e_1, e_2, e_3	exact modulus
Nil^3 β_-	$(\frac{1}{2}, -\frac{1}{2}, 0)$	exact modulus
Nil^3 β_+/u	$(-\frac{1}{2}, -\frac{1}{2}, 1)$	curvature wall
Sol^3 β_x	$(\frac{1}{3}, -\frac{1}{6}, -\frac{1}{6})$	curvature wall

Table F.1: Finite-deformation branch directions and their classification.

Branch directions (Milnor frame; tangent alignment confirmed).

Closed-form branch formula for β_- (basis of the exact modulus). On the background-adapted static Palatini branch the scale factors can be written

$$q_1(s) = q_0\, e^{s/2}, \qquad q_2(s) = q_0\, e^{-s/2}, \qquad q_3(s) = q_0, \tag{30}$$

(s is the finite-deformation parameter along β_- and q_0 a reference scale). The reduced EH density $\Delta(s)$ obtained after solving the connection fluctuations by the field equations vanishes identically at all orders in s (consistent with the residual 0 up to fourth order). For β_+/u the first-order coefficient (tadpole) of Δ is non-zero, and for β_x one of the linear/quadratic coefficients is non-zero.

Object	Check	Result
T^3 (3 directions)	$\Delta(\beta)$ symbolic up to fourth order	identically 0 (exact flat moduli)
Nil^3 β_-	above + linear-system rank check (rank = aug rank = 18, uniquely solvable)	residual 0 at all orders (exact modulus)
Nil^3 β_+/u	as above	curvature-induced non-zero term + tadpole (curvature wall; no stationary finite branch)
Sol^3 β_x	as above	non-zero (curvature wall; stationary branch absent)
S^3	—	not applicable ($d_{\mathrm{zero}} = 0$)

Table F.2: Energy-density expansion results by branch direction.

`script/scripts/paper07/finite_fate_branch_witness.py` independently reproduces the above directions and classification, within the finite Palatini branch / homogeneous minisuperspace scope. The result is based on the final branch construction.

F.4 Minisuperspace closure verification

DBR closure of the lever-complete ansatz (log-scale variables l_a, conjugate momenta p_a, lapse N):

- closure including the Hamiltonian constraint holds (lapse elimination after closure);
- the trace-free tangent norm of the DeWitt metric is positive (minisuperspace no-ghost);
- the configuration degrees of freedom of the homogeneous anisotropy = 2.

F.5 Scope and limitations (finite-fate sector)

- The walls are established within the finite Palatini branch. This is not called a full nonlinear lift/mass theorem.
- The energy-density-expansion classification is not identified with a Hamiltonian-constraint theorem / stationarity decision.
- The finite moduli count (T^3:3, Nil^3:1, Sol^3:0) is not identified with d_{zero} (3, 2, 1).
- The minisuperspace no-ghost is not reread as a full field-theory no-ghost.

G Torsion auxiliary exactness and field-promotion verification

This appendix collects the verification details of Section 8. The scope is the NY-MAIN route and the AX/VT/MX torsion ansatz scalars.

G.1 Polynomial-degree verification (exact degree-2)

Organising the reduced torsion Lagrangian into the form of Section 8.2,

$$\mathcal{L}_{\text{aux}} = a_\eta\,\eta^2 + a_V\,V^2 + b_V\,\dot{\text{vol}}\,V + (\text{boundary}), \tag{31}$$

we expanded it symbolically as a polynomial in (η, V) up to degree $D_{\max} = 4$. Since all coefficients of degree ≥ 3 are identically 0 (residual zero), it is exactly at most quadratic in (η, V).

The Euler–Lagrange equations give algebraically $\eta^* = 0$, $V^* = 0$ (the linear term in V separates to the boundary as $2\,\mathrm{d}_t(\text{vol}\cdot V)$), an *exact auxiliary shell* with no truncation. The auxiliary Hessian (the torsion mass datum) is non-negative, consistent with the stability of the shell.

G.2 Field-promotion derivative-coupling classification

For the promotion $\eta(t) \to \eta(x)$, $V(t) \to V(x)$, we classified the derivative couplings arising in the EC+NY action by sector (EC/NY) × topology (4) × mode (AX/VT/MX) × derivative type (time/spatial/mixed) × degree (128 cells in total). The results:

Derivative type	Quadratic (kinetic/gradient)	Linear
Time $(\dot\eta, \dot V)$	**absent (all cells)**	boundary-type current only: EC $2\,\text{vol}\cdot\dot V$ / NY $6\,\text{vol}^{2/3}\,\dot\eta$
Spatial $(\partial_i\eta, \partial_i V)$	**absent (all cells)**	**even a linear term absent (all cells)**
Mixed	absent	absent

Table G.1: Field-promotion derivative-coupling classification.

No quadratic derivative coupling appears at all.

Structural reason. With $\Gamma = \Gamma_{\text{LC}} + K(\eta, V)$ and $R[\Gamma] = \mathrm{d}\Gamma + \Gamma \wedge \Gamma$: if K is algebraic in η, V then dK is first order in the field derivative, and in the scalar contraction of ΔR the quadratic derivative couplings cancel structurally (Section 8.3). At the level of the

Riemann components (24 components) individual derivative terms fire, but they cancel under the scalar contraction, as confirmed by a symbolic computation of the component sum.

Cross-check. The same classification procedure applied to a free-scalar toy Lagrangian was confirmed to detect the kinetic term correctly. Reproduction: `script/scripts/paper07/torsion_auxiliary_field_promotion_witness.py` (no claim about a full 24-component torsion field theory).

G.3 Minimal definition of the NY boundary three-form (precision)

The NY four-form, in our convention, has a boundary three-form Q_{NY} satisfying

$$\mathrm{d}Q_{\mathrm{NY}} = e_I \wedge e_J \wedge R^{IJ} - T^I \wedge T_I \tag{32}$$

(this identity is confirmed to have residual zero under our convention). The NY-side time-linear current $6\,\mathrm{vol}^{2/3}\dot{\eta}$ of G.2 equals the time derivative of the Q_{NY} primitive, consistent with its boundary nature. Q_{NY} is used only within this minimal definition and is not introduced as an object of the main text (the physics of boundary charge is outside the scope of this paper).

G.4 Corollary (finite-fate invariance)

Since $\mathcal{L}_{\mathrm{aux}}|_{\mathrm{shell}} = 0$ on the auxiliary shell, the modulus/wall decision of Appendix F is invariant under all of EH/AX/VT/MX (a corollary; not an independent dynamical check).

G.5 Scope and limitations (torsion sector)

- This decision is for the field promotion of the AX/VT/MX ansatz scalars, not a $(\partial T)^2$-absence theorem for a full 24-component torsion field theory.
- A linear boundary current is not called a kinetic/propagating mode.
- The exact-degree-2 / exact-shell claim is restricted to the NY-MAIN route (no automatic extension to other routes).
- No torsional-charge theorem / quantised charge / localisation is claimed.
- The homogeneous auxiliary persistence (G.1) and the field-promotion verification (G.2) are separate layers; the latter does not reprove the former (it is an extension).

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